

Exploratory Data Analysis (EDA) Summary Report Template

1. Introduction

This report presents the findings of the exploratory data analysis (EDA) conducted on Geldium's customer dataset, as part of the Tata iQ engagement to develop an AI-powered credit delinquency prediction and intervention system. The objective of this analysis is to assess the dataset's structure, completeness, and reliability ahead of predictive modeling, and to surface initial risk indicators that can inform Geldium's Collections team strategy. GenAI tools (used in line with Tata iQ's data confidentiality standards — no raw customer-level data was shared with external tools) were used alongside standard statistical methods to accelerate pattern discovery, with all suggested treatments validated against domain knowledge before adoption.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key dataset attributes:

- Number of records: 500 customers, 19 variables
- Key variables: Customer_ID, Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Delinquent_Account (target variable), Loan_Balance, Debt_to_Income_Ratio, Employment_Status, Account_Tenure, Credit_Card_Type, Location, and six months of payment status history (Month_1–Month_6: On-time / Late / Missed)
- Data types: 8 numerical fields (Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure), 1 binary target (Delinquent_Account), 4 categorical fields (Employment_Status, Credit_Card_Type, Location, Customer_ID), and 6 categorical monthly payment-status fields

Data quality issues identified:

- No duplicate Customer_IDs or fully duplicate rows were found.

- Inconsistent category labels: Employment_Status contains six raw values that collapse to four true categories — “Employed,” “employed,” and “EMP” all represent the same status (240 records combined) and require standardization before modeling.
- Logical/range anomaly: 4 customers have a Credit_Utilization ratio above 1.0 (i.e., above 100%, up to 102.6%), which is technically possible (over-limit balances) but should be verified with Geldium as it may also indicate a data entry or calculation error.
- No negative values found in Income, Loan_Balance, or Account_Tenure — a good sign of basic data integrity.
- Credit_Score values fall within a plausible 301–847 range; 2 records have missing scores.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- Variables with missing values: Income (39 missing, 7.8%), Loan_Balance (29 missing, 5.8%), Credit_Score (2 missing, 0.4%)
- Missing data treatment: Median imputation for Income and Loan_Balance (both are right-skewed financial variables where median is more robust than mean); Credit_Score imputed using regression based on Age and Account_Tenure given the very low missingness rate

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

- Correlations observed between key variables: Surprisingly, none of the standard financial risk indicators in this dataset (Income, Credit_Score, Credit_Utilization, Debt-to-Income Ratio, Missed_Payments, Loan_Balance, Account_Tenure) show a meaningful linear correlation with the Delinquent_Account target — all correlation coefficients fall between -0.04 and +0.05. This is much weaker than industry expectations, where Credit_Utilization and Missed_Payments typically show strong association with delinquency.
- Unexpected anomalies: 61 customers with a high Credit_Score (>750) also have 3+ Missed_Payments recorded — a pattern that doesn't align with typical credit risk behavior and warrants investigation into whether Missed_Payments and Credit_Score are sourced from different systems or time periods. Delinquency rates also do not vary

meaningfully by Employment_Status, Credit_Card_Type, or Age group, which is atypical for this kind of dataset.

Items flagged for analytics team follow-up:

- The weak correlation pattern extends to the Month_1–Month_6 payment history as well: even the count of ‘Missed’ or ‘Late’ statuses across the 6-month window shows virtually no correlation with Delinquent_Account (coefficients near 0.00–0.02). This strongly suggests the dataset's target labels may have been randomly assigned for training/exercise purposes, rather than reflecting a real causal relationship between these features and actual repayment behavior.
- Recommend re-confirming the target variable definition and label source with Geldium — if Delinquent_Account is meant to reflect real customer behavior, the absence of correlation with any input feature (including payment history) is a red flag that should be resolved before proceeding to model development.
- Recommend validating the Credit_Utilization values above 100% with Geldium's source systems.
- Recommend standardizing Employment_Status labels (EMP / employed / Employed → Employed) before any modeling or segmentation work.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

Example AI prompts used:

- “Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.”
- “Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning.”
- “Suggest an imputation strategy for missing values in this dataset based on industry best practices.”
- “Analyze the correlation between customer financial variables and delinquency risk, summarizing key findings in simple terms.”

Note on data confidentiality: GenAI tools were used only for guidance on analytical approach, summarization techniques, and interpretation of aggregate statistics already computed locally. No raw, customer-identifiable data (e.g., Customer_ID-linked records) was shared with any

external GenAI tool, in line with Tata iQ's data confidentiality standards for financial services clients.

6. Conclusion & Next Steps

This EDA confirms that Geldium's dataset is structurally sound — well-formed, with low missingness (under 8% across all affected fields) and no duplicate records — making it usable for predictive modeling once minor cleanup steps (category standardization, missing value imputation, utilization outlier verification) are completed. However, the most significant finding is that none of the standard risk indicators (Income, Credit_Score, Credit_Utilization, Debt-to-Income Ratio, Missed_Payments, Loan_Balance, Account_Tenure, Employment_Status, Age, monthly payment history) show meaningful correlation with the Delinquent_Account target as currently defined. This is an important finding to flag to Geldium and the wider Analytics team before predictive modeling proceeds, since building a model on these features without resolving this issue would likely yield poor or misleading predictive performance.

Recommended next steps:

- Confirm with Geldium how the Delinquent_Account label was generated and over what time window, to rule out a labeling or data-linkage issue.
- Standardize categorical fields (Employment_Status) and resolve the Credit_Utilization outliers (>100%) with the source team.
- Apply median imputation to Income and Loan_Balance, and regression-based imputation to Credit_Score; document the rationale in the model documentation for auditability.
- Once the target variable is validated, re-run correlation and feature-importance analysis before proceeding to model selection (e.g., logistic regression as an interpretable baseline).
- Incorporate fairness checks (e.g., disparate impact across Age, Employment_Status, Location) as part of the next modeling phase, per responsible AI guardrails.